Chapter 6, Problem 15P

Problem

Two capacitors (25 and 75 μF) are connected to a 100-V source. Find the energy stored in each capacitor if they are connected in:

(a) parallel

(b) series

Step-by-step solution

Step 1 of 7

Given two capacitors are $C_1 = 25\ \mu\text{F}$ and $C_2 = 75\ \mu\text{F}$, voltage source is $v = 100\ \text{V}$.

(a)

If both capacitors are in parallel condition circuit is shown in figure (1):

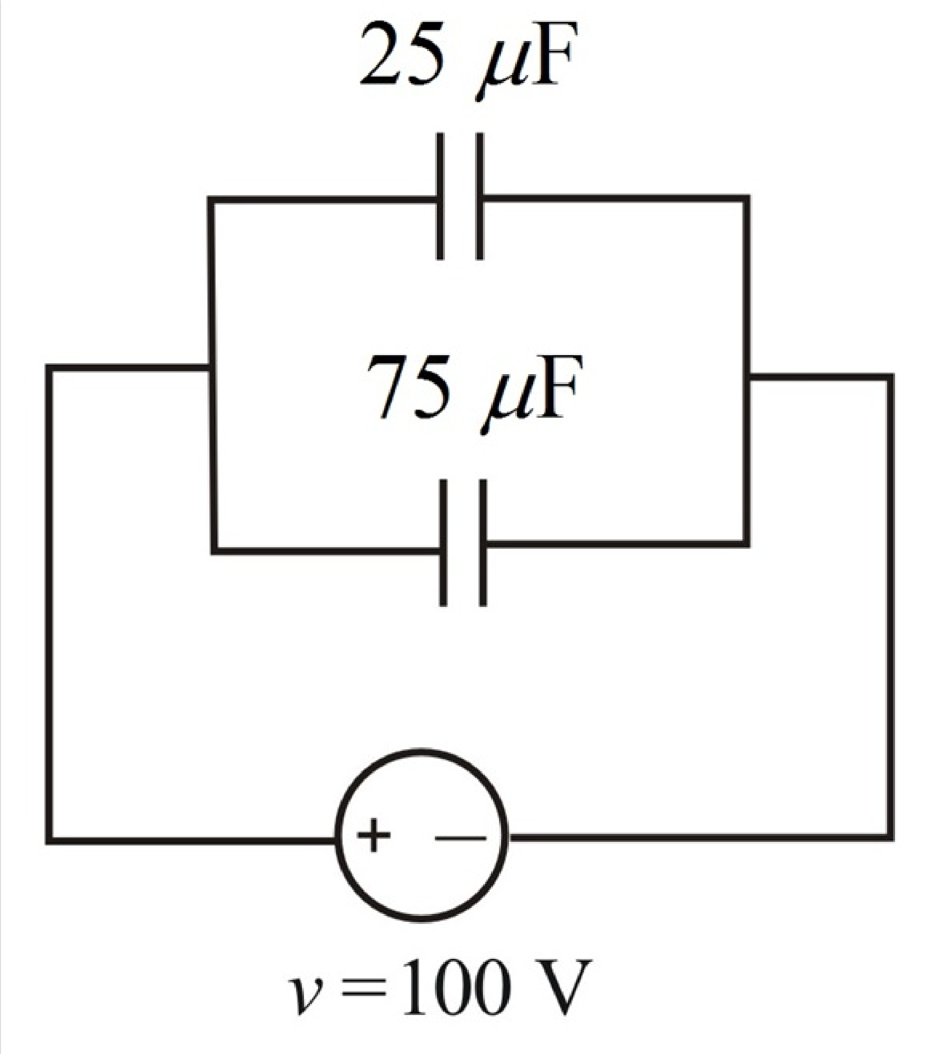

Figure (1)

Step 2 of 7

As the capacitors are in parallel the voltage across them is same as source.

Calculate the value of the energy stored in capacitor $C_1 = 25\ \mu\text{F}$.

Write the expression for energy stored in capacitor $C_1 = 25\ \mu\text{F}$.

$$\begin{aligned} w_1 &= \frac{1}{2} C_1 v^2 \\ &= \frac{1}{2} \times 25 \times 10^{-6} (100)^2 \\ &= 0.125\ \text{J} \end{aligned}$$

Hence the energy stored in capacitor $C_1 = 25\ \mu\text{F}$ is $\boxed{w_1 = 0.125\ \text{J}}$.

Step 3 of 7

As the capacitors are in parallel the voltage across them is same as source

Calculate the value of the energy stored in capacitor $C_2 = 75\ \mu\text{F}$.

Write the expression for energy stored in capacitor $C_2 = 75\ \mu\text{F}$.

$$\begin{aligned} w_2 &= \frac{1}{2} C_2 v^2 \\ &= \frac{1}{2} \times 75 \times 10^{-6} (100)^2 \\ &= 0.375\ \text{J} \end{aligned}$$

Hence the energy stored in capacitor $C_2 = 75\ \mu\text{F}$ is $\boxed{w_2 = 0.375\ \text{J}}$

Step 4 of 7

(b)

If both the capacitors are in series combination then the circuit is as shown in figure (2):

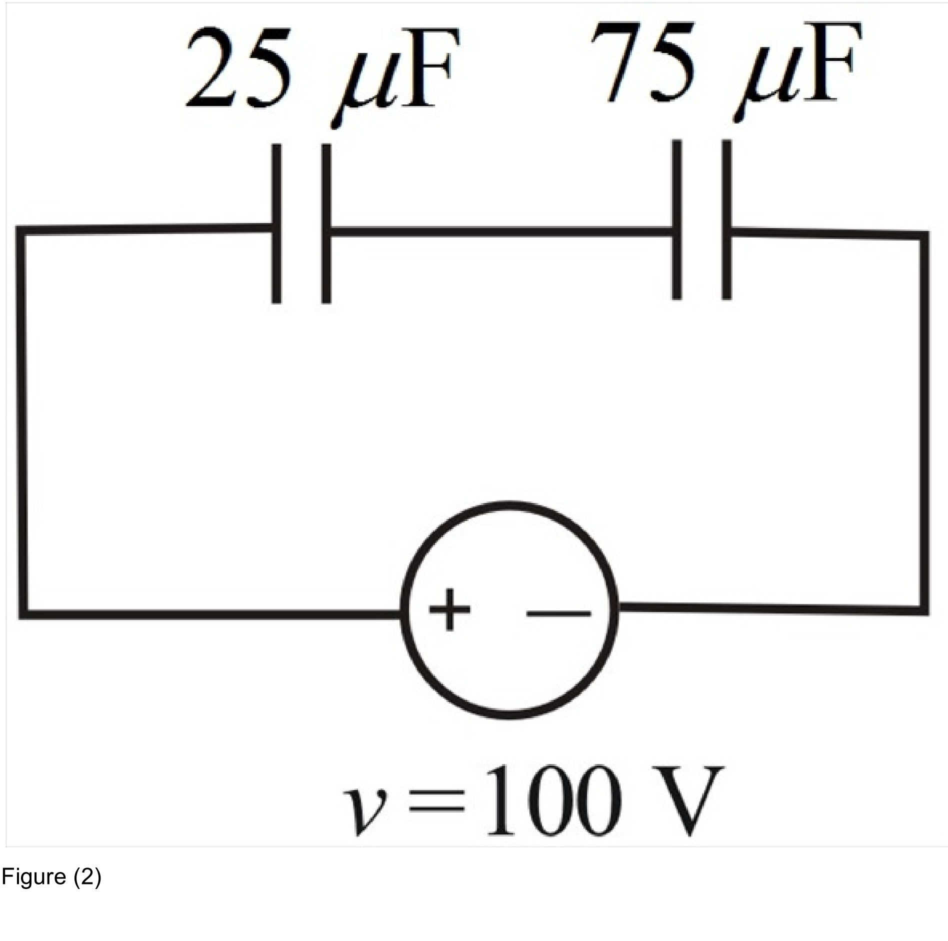

Figure (2)

Step 5 of 7

Calculate the value of equivalent capacitance C_{eq} of two capacitors.

$$\begin{aligned} C_{eq} &= \frac{C_1 C_2}{C_1 + C_2} \\ &= \frac{25 \times 75}{25 + 75} \\ &= 18.75\ \mu\text{F} \end{aligned}$$

Total charge on series capacitance circuit is q.

$$\begin{aligned} q &= C_{eq} v \\ &= 18.75 \times 100 \times 10^{-6} \\ &= 1875\ \mu C \end{aligned}$$

Calculate the value of the voltage across capacitor $C_1 = 25\ \mu\text{F}$.

$$\begin{aligned} v_1 &= \frac{q}{C_1} \\ &= \frac{18.75 \times 10^{-6}}{25 \times 10^{-6}} \\ &= 75 \end{aligned}$$

Therefore, the value of voltage across capacitor $C_1 = 25\ \mu\text{F}$ is $75\ \text{V}$.

Calculate the value of the voltage across capacitor $C_2 = 75\ \mu\text{F}$.

$$\begin{aligned} v_2 &= \frac{q}{C_2} \\ &= \frac{18.75 \times 10^{-6}}{75 \times 10^{-6}} \\ &= 25 \end{aligned}$$

Therefore, the value of voltage across capacitor $C_2 = 75\ \mu\text{F}$ is $25\ \text{V}$.

Step 6 of 7

Calculate the value of the energy w_1 stored in capacitor $C_1 = 25\ \mu\text{F}$.

$$\begin{aligned} w_1 &= \frac{1}{2} C_1 v_1^2 \\ &= \frac{1}{2} \times 25 \times 10^{-6} (75)^2 \\ &= 0.070\ \text{J} \end{aligned}$$

Hence the energy w_1 stored in capacitor $C_1 = 25\ \mu\text{F}$ is $\boxed{0.070\ \text{J}}$

Step 7 of 7

Calculate the value of the energy w_2 stored in capacitor $C_2 = 75\ \mu\text{F}$.

$$\begin{aligned} w_2 &= \frac{1}{2} C_2 v_2^2 \\ &= \frac{1}{2} \times 75 \times 10^{-6} (25)^2 \\ &= 0.023\ \text{J} \end{aligned}$$

Hence the energy w_2 stored in capacitor $C_2 = 75\ \mu\text{F}$ is $\boxed{0.023\ \text{J}}$.